Assignment

Module 2.2 Power Supply Systems Theory

1. **List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.**

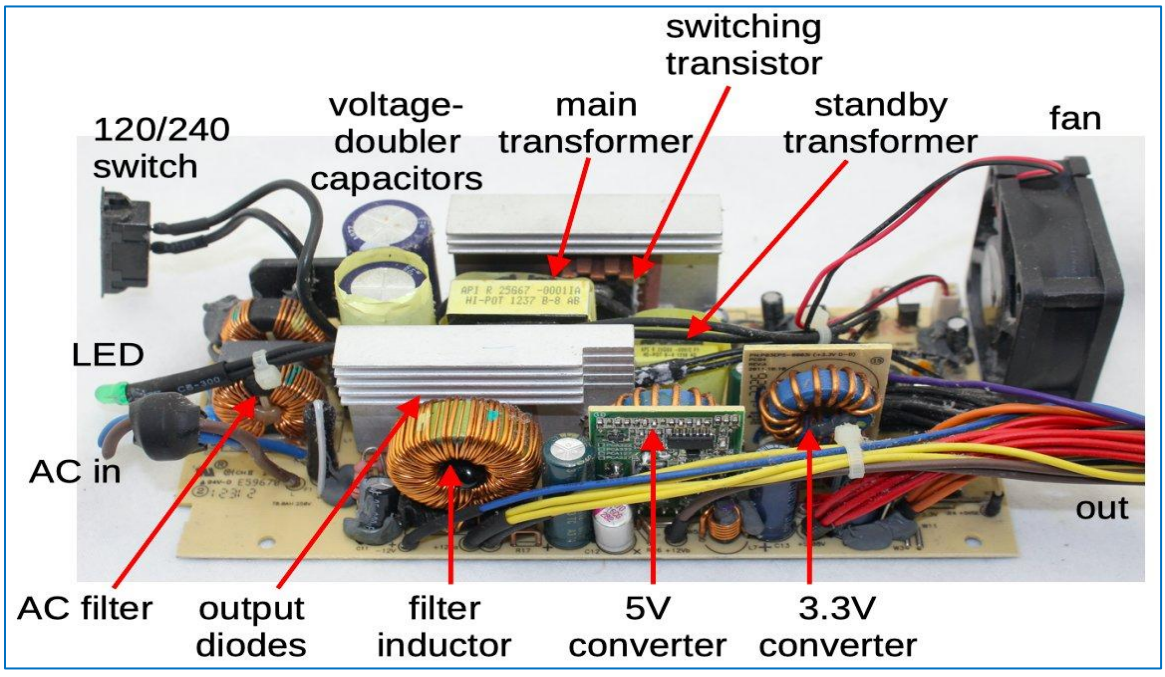
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Type	Device	Reason
SMPS	Desktop PC	Converts AC electricity into the required regulated DC power for the computer components.
SMPS	LED TV	Supplies stable and efficient DC power to the television's electronic circuits.
SMPS	Wi-Fi Router	Converts household AC power into low-voltage DC power required for the router.
UPS	Desktop Computer	Provides backup power during outages and prevents sudden shutdowns.
UPS	CCTV System	Maintains continuous operation of cameras and recording equipment during power failures.

2. **Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.

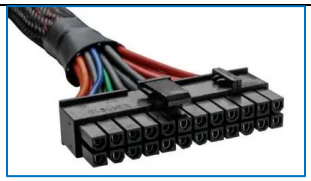
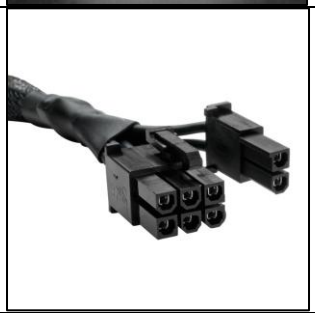
Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.**

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3. **Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.**

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No.	Connector Name	Image	Uses	Safety Tip
1	ATX 24-pin motherboard connector		Provides the main power supply to the motherboard.	Turn off and unplug the computer before connecting or removing it.
2	SATA power connector		Supplies power to SATA hard drives, SSDs, and optical drives.	Align the connector correctly and never force it into the socket.
3	Molex 4-pin connector		Powers older hard drives, optical drives, and some cooling fans.	Hold the plastic plug when removing it; do not pull the wires.
4	PCIe 6/8-pin connector		Provides additional power to compatible graphics cards.	Use the correct PCIe power cable to ensure it is securely connected.

4. **Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.**

➢ how to connect SMPS to the motherboard:

Step 1: Put the power supply in Place the power supply inside the computer case and secure it with screws.
Step 2: Connect the main board Plug the 24-pin ATX cable from the power supply into the main power port on the motherboard.
Step 3: Connect the CPU. Attach the 4-pin or 8-pin CPU power cable into the CPU power port on the motherboard.
Step 4: Connect the video card Plug the 6-pin or 8-pin PCIe power cable, or the 12V-2x6 cable, into the graphics card.
Step 5: Connect hard drives and SSDs Use the 15-pin SATA power cable to power your hard drives (HDDs) and solid-state drives (SSDs).
Step 6: Connect other parts Use a 4-pin Molex cable for older drives, fans, or other devices that need it.
Step 7: Check all connections. Make sure every cable is tight and doesn't block the cooling fans.